first strategy in 2023, as they look to increase agility and efficiency while reducing costs.

Cloud adoption leads to:

- An expanding application ecosystem, which drives market responsiveness
- Changing business model – enterprises soon become integrators of best-of-breed services through collaboration
- New regulatory requirements driven by a global collaborative economy and the need to address open markets
- Better customer experience
- Transformation and optimisation across different process stacks: sales, front office, middle office and back office

The top tech trends as a direct result of cloud adoption are:

- Data centre rationalisation, as hybrid clouds replace data centres.
- Edge computing — data is stored and processed at the edge of the network, and analysed geographically closer to its source.
- Increased automation and self-learning capabilities, greater data security and privacy, and more personalised cloud experiences, all due to AI and ML.
- Disaster recovery — cloud computing is effective in disaster recovery and offers businesses the ability to quickly restore critical systems in the event of a natural or man-made catastrophe.
- Multi- and hybrid clouds offer a combination of public and private clouds dedicated to a specific company, the data of which is the key business driver, such as insurance companies, banks, etc.
- Kubernetes enables large scale deployments that are highly scalable and efficient. It is an extensible, open source platform that runs applications from a single source while centrally managing services and workloads.
- Movement of IT development and testing to the cloud.
- Maximised productivity with scalability and high availability. Core functions will increasingly move to private cloud and non-core to public cloud.
- Emergence of cloud service brokerages – move towards a hybrid model.
- Enterprises moving beyond traditional roles by offering new digitised products like cloud based storage for customer files.

Let's see how different industries are making use of the cloud.

- Communications (telecom services providing point-to-point contact): Converged billing, multi-party settlement, financials, HR, procurement
- Governments (federal, state and local government agencies as well as non-profits): Public safety software, grants and performance management, computer assisted dispatch, jail management, courts management, tax collections, records management, financials, HR, procurement
- Healthcare (healthcare providers, hospitals, physicians, clinics, hospices): Electronic health records, practice management, revenue cycle management, financials, HR, procurement
- Chemicals (fuel extraction, petroleum refining, primary derivatives from oil, chemicals and allied products): Oil and gas operations management, reserves management, capital planning and budgeting, well production software, financials, HR, procurement
- Utilities (electricity, water and gas utilities): Customer care, billing, smart meter infrastructure, energy trading risk management, SCADA, financials, HR, procurement

Drivers for open source cloud adoption

Generally, cloud computing must satisfy five essential principles. These are: on-demand service, access network, resource pooling, elasticity, and measured services. To do this, cloud computing provides three kinds of service models: Software as a Service (SaaS), Platform as a Service (PaaS) and Infrastructure as a Service.

By interacting with various customers, I have arrived at the following key reasons for which a business moves to the cloud:

- Reduction of capex and opex to deliver business services
- Minimisation of IT costs by improving delivery time and quality of the application
- Businesses can move applications to the cloud at their own speed owing to different factors like complexity of the application, data requirements, regulatory and compliance needs, modernisation prerequisites, cost implications, real-time requirements, etc
- Best-of-breed environments can be selected for the applications, using special hardware to run and deliver the functionality at the most optimal cost and effort
- Security and regulatory requirements — applications requiring jurisdictional provisioning and regulatory compliance require deeper due diligence to select the right candidates for cloud and the right hosting option for the application
- Complex integrations — applications with high integration may need to be co-located to deliver the best experience. Point of delivery applications can be deployed on specialised hardware systems or integrated with machinery that requires these applications to be hosted on-premises
- Seasonal requirements — applications that have seasonal spikes can employ cloud services to handle the additional load for a short period and avoid capital investments
- High availability — applications can be redundantly deployed for